

Rafsan Mahmud

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Research Interests

Robust and Trustworthy Deep Learning; Explainable AI; Generative Models; Computer Vision; Adversarial Machine Learning; Anomaly Detection; Medical Image Analysis.

Education

American International University–Bangladesh (AIUB), Dhaka, Bangladesh Feb 2022 – Apr 2026
Bachelor of Science in Computer Science & Engineering
CGPA: 3.98/4.00

Undergraduate Thesis: *Latent-Space Anomaly Detection via Explainable AI for Enhancing GAN Robustness*
Supervisor: Dr. Debajyoti Karmaker, Associate Professor, AIUB

Research Experience

[A Placeholder For Position] Aug 2026 – Present
ELITE Lab AI

- **Research Focus:** [Placeholder for core focus area]
- **Research Activities:** [Placeholder for methods and pipeline implementation]
- **Research Contribution:** [Placeholder for key technical innovation]
- **Research Outcomes:** [Placeholder for results and manuscripts]

Undergraduate Researcher Dec 2024 – April 2026
American International University–Bangladesh (AIUB), Dhaka, Bangladesh

Latent-Space Anomaly Detection via Explainable AI for Enhancing GAN Robustness

- Developed an explainable anomaly detection framework for identifying adversarial perturbations in GAN latent spaces.
- Proposed a dual-autoencoder approach combining a deterministic LatentExplainer and a variational LatentVAE to model deviations from the learned latent manifold.
- Investigated the proposed framework across WGAN-GP and StyleGAN2 architectures using MNIST, CIFAR-10, and CelebA datasets.
- Evaluated the framework under adversarial attacks including FGSM, PGD, and CW.
- Investigated latent-space representations and explainability as mechanisms for detecting adversarial perturbations in generative models.

Manuscripts in Preparation

- **Mahmud, R.,** Mahbuba, S., Alam, T., & Islam, M. B. *Monitoring Model Degradation Under Progressive Image Corruption: Early-Warning Behavioral Signals and Explanation Instability*. Conference manuscript in preparation.
- **Mahmud, R.,** Mahmud, A., Emona, I. B., & Shajid, M. S. A. *Latent-Space Anomaly Detection via Explainable AI for Enhancing GAN Robustness*. Manuscript in preparation.
- **Mahmud, R.,** & Mahmud, A. *Comparative Analysis of U-Net Variants for Brain Tumor Segmentation*. Manuscript in preparation.

Awards & Honors

Summa Cum Laude – Awarded for outstanding academic achievement under the honors program. Dec 2026
Dean's Honor Award – Recipient of six Dean's Honor Awards for outstanding academic performance. Dec 2022 – Apr 2026
Merit Scholarship – Awarded for consistent academic excellence during undergraduate studies. Dec 2022 – Apr 2026
Talentpool Scholarship, JSC – Awarded for performance in Junior School Certificate examination. Mar 2017

Research Projects

Monitoring Model Degradation Under Progressive Image Corruption2026

PyTorch · CIFAR-10-C · Explainable AI · Grad-CAM

- Evaluated an ImageNet-pretrained ResNet-50 on CIFAR-10-C across 15 corruption types and five severity levels to characterize model degradation under progressive image corruption.
- Analyzed confidence change, Jensen–Shannon divergence, and prediction flip rate as behavioral signals for detecting emerging performance degradation.
- Demonstrated that behavioral instability provided early warning for 9 of 15 corruption types, with a mean lead of 2 severity levels before accuracy fell below 80%.

Comparative Analysis of U-Net Variants for Brain Tumor Segmentation2025

TensorFlow · PyTorch · Medical Imaging · BraTS20

- Conducted a comparative study of 3D U-Net, Residual 3D U-Net, and U-Net 3+ for brain MRI segmentation.
- Evaluated model performance using Dice, Jaccard, and mIoU metrics.
- Found U-Net 3+ to achieve the strongest segmentation performance among evaluated architectures.

Real-Time Face Detection & Recognition with CNN2025

Python · TensorFlow · OpenCV · Deep Learning

- Developed and trained a custom convolutional neural network using 4,000+ facial images for real-time face recognition.
- Implemented webcam-based inference using OpenCV and optimized frame processing for real-time operation.

Technical Skills

Programming: Python, C++, C, C#, Java, JavaScript, PHP
Machine Learning & AI: Deep Learning, Computer Vision, Generative Models, Explainable AI, Anomaly Detection, Image Segmentation, Object Detection, Federated Learning
Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, YOLOv8, U-Net, U-Net 3+, Flower, .NET, ASP.NET
Research & Computing: Google Colab, Model Evaluation, Adversarial Machine Learning, Medical Image Analysis
Other Technologies: MySQL, SQL Server, Selenium, AJAX, jQuery

Professional Experience

Quality Assurance Engineer InternJune 2025 – October 2025

Deepchain Labs, Dhaka, Bangladesh

- Performed manual testing of web and mobile applications to identify functional defects and usability issues.
- Designed and executed functional and regression test cases across multiple product modules.
- Prepared detailed bug reports and collaborated with developers to facilitate timely resolution.
- Documented product requirements, testing workflows, and quality assurance processes.

Professional Development

Software Testing & Quality AssuranceNov 2024 – Jan 2025

Institute of Information Technology (IIT), Jahangirnagar University – Certified 80-hour program under EDGE Project

IT Essentials: PC Hardware and SoftwareDec 2023

American International University–Bangladesh (Cisco Networking Academy)

Language Proficiency

IELTS Academic: Overall Band: [Score] (L: [X], R: [X], W: [X], S: [X])

Target: [Month Year]

References

Dr. Debajyoti Karmaker Sessional Academic School of Computing Tech. RMIT University, Australia <i>Former Head & Assoc. Prof., AIUB Thesis Supervisor</i> debajyoti.karmaker@rmit.edu.au Scholar Profile	Md. Kishor Morol AI Safety Researcher, Meta Founder & CEO, ELITE Lab New York, United States <i>Former Asst. Professor, AIUB Research Director / PI</i> kishormorol@ieee.org Scholar Profile	Dr. M. Jashim Uddin Professor & Convener Department of Mathematics AIUB, Dhaka, Bangladesh <i>Former Head, Mathematics, AIUB Academic / Course Instructor</i> mdjashim@aiub.edu Scholar Profile
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